



DEALER BEST PRACTICE SERIES

Pneumatic Service Pit
Barrier

Maintenance and Repair Application Component Life Management Component Renewal (CRC) MARC Management

Pneumatic Service Pit Barrier0

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1.0 Introduction

Due to limited ground clearances, service pits are now a common practice with underground mining workshops. Upon completion of service, the pits can present a potential falling hazard. To mitigate this risk, sites commonly use barriers or plates. Either approach involves heavy lifting or use of forklifts.

This Best Practice details a pneumatically operated barrier that moves into place and retracts for service with a simple switch. This system provides a safety barrier to the pit without heavy lifting and also saves time required with construction.

2.0 Best Practice Description

The barrier is divided into two sections. The proportion of the two sections can be adjusted per site requirements. However, the shorter section needs to be long enough to have two hinged points to properly leverage it in and out of the way. One section must have the intersecting portion made narrower to fit within the opposite section when collapsed into the pit; to avoid an interference of the two sections. Image 1 shows the barrier in the erected position.



Image 1: Pit barrier in the erected position

Both sections are operated by dedicated pneumatic actuators. They can also be plumbed to operate independently if required. When the barrier is folded down, the sections sit below the floor level to limit obstruction. This prevents the barrier itself from being damaged by parts of the machines moving over the pit. Image 2 shows the barrier already folded down.

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Fig 2: Pit barrier in folded position

3.0 Implementation Steps

This specific pit barrier was procured through Booth Engineering in Australia. The product is built to Australian and New Zealand standards. Caterpillar does not endorse this particular supplier. However, our customers had experience with this company and product. There are also barriers for other applications in working at heights.

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Safety Concerns

- Various pinch points exist with this solution. Booth Engineering has agreed to provide documentation on this upon request
- Can the barrier collapse unexpectedly?
 - o The pneumatic cylinders hold everything in raised position and lowered position. It doesn't lock in place, the air pressure is sufficient to hold it very firmly in place.
 - o The pneumatic cylinders are fitted with pilot check valves. If the air compressor is turned off overnight and pressure runs down, the pilot check valves ensure that the air is held in the bottom end of the cylinders, thereby keeping the rails raised. The pilot check valves require a positive pressure to allow them to release air from the cylinders. The only way to lower them is to operate the main control valve when air pressure has been built up.

4.0 Benefits

1. Safety against personnel falling into service pits.
2. Eliminate heavy manual handling by other methods of pit barriers.
3. Reduce labor time requirement to set up and take down manual barriers.

5.0 Resources Required

Cost of installation is varied, depending on the structure of the pit lining for attachment of mounting points. Cost also depends on any required modifications such as the staircase top for the barrier to fit below and surface and also avoid tripping of personnel when using the staircase.

6.0 Supporting Attachments / References

None.

7.0 Related Best Practices

None at this time.

8.0 Acknowledgements

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